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A Theory Instrument for reimagining embodied practices

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Abstract: Embodied Sensemaking Theory describes how people make sense in ongoing interactions with the social and material world. It has potential in projects aimed at changing embodied practices. However, designers often find it challenging to use this complex theory. We build on recent research on tangible 'Theory Instruments' for designers. We designed a Theory Instrument for embodied sensemaking with design students who design for social interactions and with young people who investigate their energy consumption. Our analysis of 12 experimental sessions shows how Embodied Sensemaking Theory helps reimagine human practices towards more sustainable futures. Our contribution is two-fold: We show that experiential actions (e.g. weaving lines, shaping textiles, wearing bodybands), rather than the tangible things as such, can represent theory key-aspects in use. We develop a logic of how to disentangle the complexity of lifeworld, socially situated practices, skills and affordances, action-perception couplings, rules and signs.

Keywords: embodied practices, design process, tangible interaction, young people

1. Introduction

Theoretical perspectives can enable challenging unquestioned practices, and in relation to issues such as sustainability can point towards new potentials. We investigate how *Embodied Sensemaking Theory* can be utilised in design work in a tangible way. This is exploratory research: We co-develop a Theory Instrument with graduate design students and explore how our Embodied Sensemaking Instrument (Figure 1) scaffolds young people in reimagining the embodied nature of their daily energy consumption and in ideating solutions for more sustainable habits.

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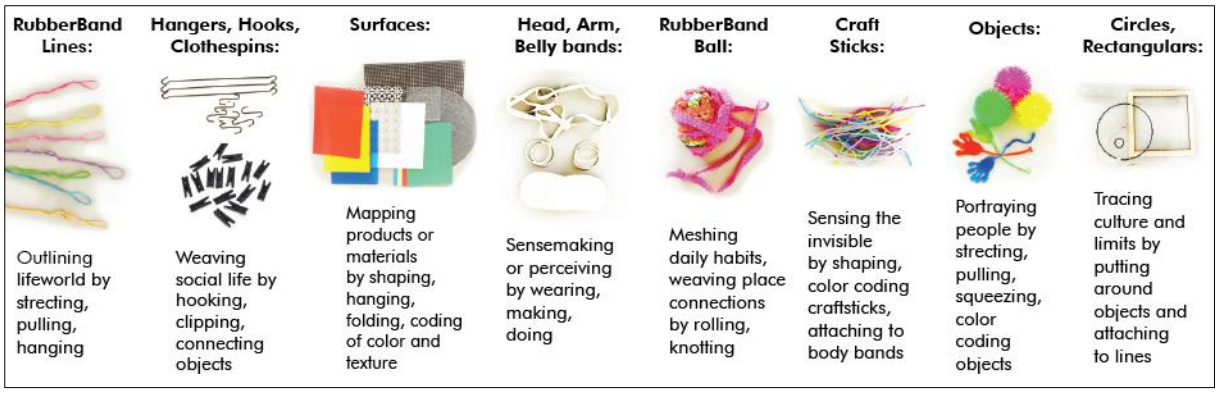


Figure 1 Components of the Embodied Sensemaking Instrument that participants employ to outline lifeworld, socially situated practices, skills, and affordances, norms, rules, etc.

We build on the concept of 'Theory Instruments' developed by Buur et al. (2022). The Theory Instruments created so far could be characterised as sets of objects which represent the main aspects of a given theory which can be helpful to inspire a project team to approach a problem in a novel way that is informed by the theory. Designing an Embodied Sensemaking Theory Instrument has proved unique: The theory itself argues that in human sensemaking objects do not only represent meanings, but they are involved in the emergent, embodied experience through which human beings literally make sense in action. The implication is that beyond relying on project members to build a physical shared representation of an issue at hand, such a theory instrument allows project members to actively explore the theory and make sense of interaction.

2. Background

2.1. Theories of embodied sensemaking

Theories of embodiment have been influential in design literature such as interaction design for more than two decades (e.g. Dourish, 2001; Svanæs 2001). The phenomenological concept of embodiment underlines the fact that human experience relies on the body's active and skillful engagement with the world (Merleau-Ponty, 1962). One core aspect of this is the coupling of perception and action, whereby intentional movements of the body are interlinked with what opportunities one perceives in the environment (Dant, 2008). The coupling of action and perception is often related to the concept of affordances, which also describes the immediate perception of action possibilities (Gibson, 1979). Yet, interactions tend to involve not merely the realisation of those possibilities but their careful, dexterous execution through skilled practice (Ingold, 2011). This understanding is further aligned with the concept of embodied cognition, which highlights the role of the human bodily experience in thinking within and making sense of the world and how thinking is entangled with making (Malafouris, 2013). From this point of view, design activity is tasked with shaping not only the functional, interactive and aesthetic qualities of its products but, more fundamentally, the embodied and situated interactions around them: Designed things mediate human experience (Verbeek, 2001).

van Dijk's (2018) paper provides a comprehensive theoretical framework for "designing for embodied being-in-the-world." According to the author, design scaffolds the ways in which people are coupled with others and things in their environment – one's Lifeworld. The visualisation in Figure 2 is particularly useful (van Dijk, 2022). One of Hummels and van Dijk's (2015) principles of embodied sensemaking is "social-situatedness," which they represent with a portable stage. The stage may be considered a "web" of lines or a "mesh world", with reference to Tim Ingold's (2015) Life of Lines research. Ingold (2015) sees social life as lines and blobs that are tangled and connected with each other. Lifeworld – the meaningful world in which one acts – collapses everything around us: our body, tools, materials, traces, creative expression, and external representation, and the interactions between all these things in the form of a mesh.

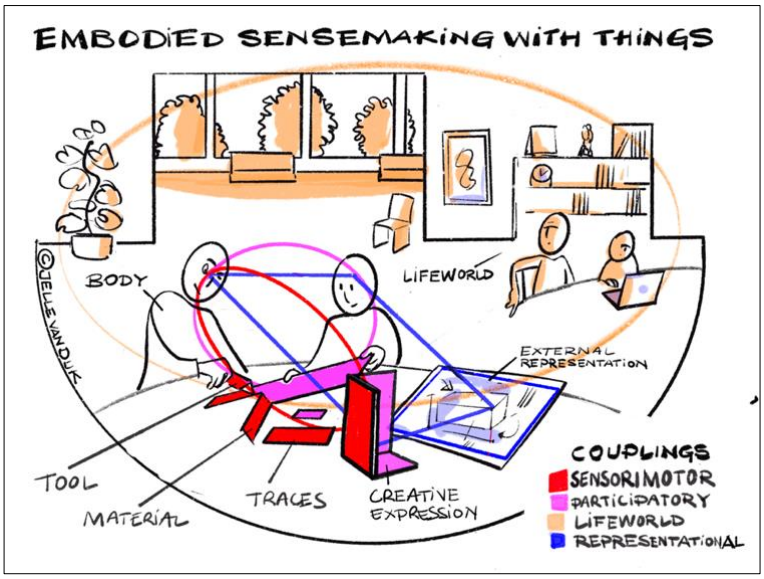


Figure 2 An illustration of the core aspects of Embodied Sensemaking (author, forthcoming).

2.2 The Idea of Theory Instruments

Theory Instruments were originally proposed as a way of boosting the application of theories from social sciences and anthropology with design practitioners in the industry (Buur et al., 2023). The 'Actor Network Rings' instrument (Agca & Buur, 2023), for instance, challenges participants to attach colourful clothespins to a wooden ring representing actors and actants in a network. To 'uphold a programme of actions' the pins must carry a wooden ball on the table. The use of physical materials to represent abstract, theoretical concepts and their relations proved highly successful with industrialists:

> "the literal tangibility of the instruments, and the metaphorical tangibility of the theories come together to create elucidating experience conducive to co-creative work. (...) The music metaphor seems to help us emphasise a role of theory not as 'solutions' to problems, but as different perspectives that can be 'played' to gain new insights into practices." (Buur et al., 2022, p.5)

Our Theory Instrument design process moved from visuals to 3D shapes, Figure 3. But unlike table-top Theory Instruments, we wanted to create an environment in which participants

could physically participate not only with their hands but with their entire bodies to create a life-sized experience.



Figure 3 The Theory Instrument design moves from a 2D theory drawing to a 3D prototype (left) and from a tabletop to a full-body experience (right).

2.3 Young people and energy

Young generations represent a large part of the world's population. In this sense, they play a crucial role in shaping the future of renewable energy with their unique understanding of consumption (Seemiller & Grace, 2016). In recent years, they have often been at the forefront of technological advancements, awareness campaigns, environmental activism, and innovative solutions to address the world's energy challenges (YESEUROPE, 2016; FFF, 2018; EYEN, 2020, etc.). Encouraging young people to develop a strong environmental awareness holds significant importance.

As energy is abstract, but consumption relates heavily to habits and norms, Embodied Sensemaking Theory would have potential for investigating energy consumption - both for young people themselves and for designers who strive to lower energy consumption.

3. Methods of experiments

We invited our participants, young people, to collaborate with us in designing the new Theory Instrument. In this process, we conducted 12 consecutive workshops in two universities. At the first university, we worked with 5 design students aged 18 to 24 years to co-design the first prototype of the instrument. At the second university, we engaged 7 professionals aged 24-35, living by themselves, in mapping their energy consumption, iteratively refining the initial prototype. At both universities, during one-hour in-person workshops, participants' consent was obtained to record video and audio. In both environments, the first author facilitated the experiments as an active participant in the study. Theoretically we understand facilitation as *"taking part in the moment to moment, improvised interactions between all, where possible outcomes are a change in the social figuration of those involved"* (Mosley & Larsen 2021), rather than as a neutral, detached role that merely assists participants in communicating their experiences, attitudes, and future expectations.

3.1. Iteration I: The case of students designing for social interaction

In a master course on Embodied Interaction design, we noticed that embodied theory is quite complex to understand for graduate design students. In the spirit of the theory itself, we decided to represent the theory in a concrete way so students might physically experience the theory.

We planned 4 in-person workshops with participants, providing them with a set of tinkering materials and rubberband lines. These workshops were conducted with 1 professional and 4 graduate design students (Table 1) who were part of the embodied interaction course. The focus of these workshops was the participants' projects, which were about how embodied interactive technologies might foster one-to-one social interaction in concrete contexts, such as first dates and project meetings. After each workshop, reflections were gathered from the students through a questionnaire (Figure 5), and the experiences obtained were used to determine the materials or general outlines for the next workshops.

Table 1 Participants and their design projects in the workshops at the first university.

P1	Professional	Negotiating House Renovation	15.06.2023
P2	Design Student	Ideating a narrative game	16.06.2023
P3-P4	Design Students	Eye-contacting between strangers	22.06.2023
P5	Design Student	Balancing office tension	29.06.2023

3.2. Iteration II: The case of young people mapping their energy consumption

Taking the insights from our first iteration, we moved to the next case environment. Considering young people's concerns about global warming and their struggles towards sustainability, this research environment is interested in understanding how young people relate to energy and in raising awareness about changing consumption behaviour.

We organised in-person workshops with 7 young people between 18-35 years old (Table 2) and gave them a choice of three different daily practices: cooking, showering, and heating. The participants worked with the prototype of the Theory Instrument produced on the basis of iteration I, and we recorded videos of both the full process and a short enactment of their instrument. At the end of each session, we recorded their reflections.

Table 2 Participants and their energy practices in the workshops at the second university.

P6	Professional	Dinner	21.08.2023
P7	Bachelor student	Morning shower	28.08.2023
P8	Professional	Planned Dinner	29.08.2023
P9	Graduate student	Short shower	05.09.2023
P10	Graduate student	Vegan dinner	15.09.2023
P11	Professional	Heaters at home	15.09.2023
P12	Graduate student	Dinner preparation	28.09.2023

4. Iteration I: Students designing for social interaction

Through the workshops of the first and second iteration we developed an understanding of how the relevant aspects of Embodied Sensemaking Theory would be represented in material actions and logical steps in 'playing the instrument'. Table 3 summarises our understanding. The left column lists those aspects of the theory that we aimed to include in the Theory Instrument. The middle column describes the logical steps that constituted a preferred order of sensemaking (outlining lifeworld, mapping skills, etc.). The right column lists those concrete actions that we co-designed with the participants: hanging, pulling, attaching the components of the instrument etc. The arrows indicate two opposed movements: For co-designing the instrument, we moved from left to right: From the theory aspects we wanted to convey, through the steps to the actions. In the analysis of video recordings, we moved from right to left: Identified the material actions, looked for logical steps, and identified theory concepts in play.

Table 3 Overview of theory aspects, logic steps, and material actions of the Theory Instrument

RELEVANT ASPECTS OF THE THEORY	LOGICAL STEPS OF THE INSTRUMENT WORKSHOP	MATERIAL ACTIONS TO EXPERIENCE THE THEORY
the lifeworld	outlining lifeworld	hanging, pulling, attaching, pointing, stretching
action perception couplings affordance skills	mapping skills	hanging, attaching, shaping, moving, throwing
socially situated practices members habits/routines tools norms and rules	mapping tools meshing routines and habits placing community tracing norms and rules	hanging, attaching, shaping, stretching, clipping, colour and texture coding
participatory sensemaking	making embodied sense	hanging, attaching, clipping, colour and texture coding wearing, scaling, setting in motion
being-in-the-world enactment phenomenology	enacting phenomenology	embedding, carrying, wearing sizing, setting in motion



4.1. The first instrument prototype

The core idea of the Theory Instrument is to materialise a Lifeworld, as depicted in Figure 2, inspired by Ingold's *Life of Lines (2015)*. The initial design concept was for the participants to build a pliable mesh or net utilising elastic bands. These “lines'' were interlinked with cable ties and metal hooks for attaching more lines. Items like wooden and plastic boards, paper boards, transparent plastic folders, plastic tiles, fabrics, stickers, balloons, pipelines, and toys were included to represent the various aspects of the theory, Figure 4.

Through four sessions, we iteratively developed the design of the instrument, the various materials, and the logical steps of ‘playing the instrument’. We first engaged a fellow professional designer (P1) to develop the basic concept and ensure relevance, and then we invited graduate design students to three consecutive sessions (P2, P3+P4, P5).



Figure 4 Rubberband lines and tinkering materials for the first instrument workshop.

Negotiating house renovation: Lifeworld as a tent

In the first session, the facilitator (F) hangs the rubberbands at the ceiling with the help of metal hangers. F outlines the theory and what the lines may represent as a social place, for instance, a room, a workplace. She asks the designer (P1) to think of a ‘design project’ to investigate. P1 steps under the lines (Figure 5.1) and suggests working with her ‘house renovation’:

P1: *A week ago, that was the weekend, I had a sudden urge to change the interior of my living room* (…) *The house is new, my boyfriend moved in with me one and a half years ago.*

F challenges P1 to elaborate on her idea using the materials.

P1: *My idea was we put a very nice table into the second part of our living room, which has direct access to the big window. My boyfriend and his family really prefer to put the big table in the corner. But there is no sunlight there. (…)*

P1 picks up a glossy surface to represent the window (Figure 5.2).

P1: I think I like the shape [of the sheet]; this is the shape of the door. (hangs it on the line; Figure 5.3) *And I have a desire to move the nice dining table towards you know the big window.* (draws a long table on a piece of paper)

When F asks if she has discussed her idea with her boyfriend, she responds:

P1: No, we didn't properly talk. Maybe this (indicates a red and blue sticky palm toy)*, and I tried to adapt myself to the norms of my boyfriend and my boyfriend's family. Those things (three different coloured circles) can represent… what should I do with them…* (Figure 5.4)

P1: (takes a craft stick) *yeah, maybe like this* (ties three circles each other) *so like this, it's the experience, life experience, for example, my boyfriend lives in a certain way, his parents have a certain way, so we have different memories and different directions towards how to live. I think for me, I like trying things, but my boyfriend doesn't. So, somehow, we integrated different ways of living.* (Figure 5.5)

P1 then builds the "lifeworld" of the new order with lines and connects them with the three circles that represent her, her boyfriend's, and his family's norms (Figure 5.6).



Figure 5 Embodied sensemaking of a house renovation in 6 instrument steps.

In this workshop, the rubberband lines come to represent the participant's house and rooms. P1 creates a tent-like structure in which she embeds herself. The most inspiring finding is how she uses objects to represent individuals and their disagreements. Here, the instrument elements represent past experiences, norms and cultural differences. Using the materials in this form did not come easy to the participant; the activity needed much facilitator interaction.

In our theoretical analysis, we abstract that P1 uses the Theory Instrument to (1-2) outline a lifeworld, (3) place her community, (4) trace norms, and possibly (5) map communication skills. We learned that, in the sessions following, we would need to focus more on the tinkering materials to find patterns in how different objects represent aspects of experience. In addition, F would offer to join the participants whenever they hesitated or needed a second role.

Soundbox for Strangers: Couplings as multiple eye contacts

F hangs the rubberband lines at the ceiling and asks the two graduate design students (P3 + P4) to wear objects (armbands, a headband and a belt) to represent sensors and connections between the things and their body. She offers simple examples of what they can do with them. P3 and P4 want to investigate their current design project by roleplaying how users might interact with their design. The product is a soundbox that helps two

complete strangers take the first step towards getting to know one another. It urges people to make eye contact when they hear a jiggling sound.

F suggests that they start by demonstrating the "before their product" story. P3 and P4 walk around as strangers (Figure 6.1). P3 takes two hangers and puts them on her ears. Then, she pulls a green line, hangs it with one hanger and pretends to listen to music. P4 takes a paper sticker as a phone and hooks himself to another line (Figure 6.2). Then they place two chairs back to back where the lines meet to represent their product (Figure 6.3). They choose two wooden rings and explain:

P3: *Maybe with this we can slide this* (takes a hook, hangs it and slides it across the line).

P4: …*and this is also making a little bit of sound* (plays with screws on the embroidery rings and makes a sound; Figure 6.4)*. If something interrupts the usual motion* (hits one of the big rings to make a sound; Figure 6.5) *and someone gets in that thing and makes a sound, then they also look at each other.*

P4: [This] connected to what my lecturer talked about as traces if I move a little bit above (moves a hook on the line: Figure 6.6), *it is kind of a different phase maybe someone else is going to see it, and set it up again.*

They mimic eye contact with different materials and actions: shaking sticky hands by throwing them at each other (Figure 6.7) or putting the same colour lines close to their eyes with the headband and hooks (Figure 6.8).

At the end of the workshop, F asks what these lines represent now.

P1: (holding all three lines together) *Now [they are] multiple eye contacts and multiple connections with multiple people* (Figure 6.9).

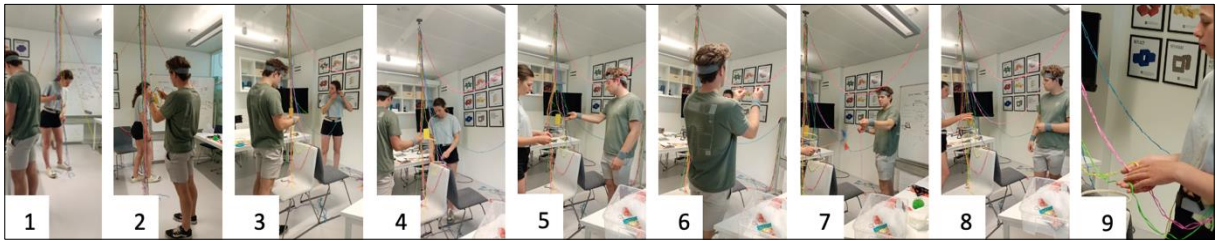


Figure 6 Embodied Sensemaking of a 'Soundbox for Strangers' design.

In this workshop, participants use the tool not only to represent but also to act out the ways in which they, as users, would interact with one another through the product they designed. This can be contrasted to P1's use of the instrument to represent norms and relationships in a more abstract fashion. Instead, P3 and P4's engagement with the tool is real-time and full-body. They either play with the materials (making sounds) or use the tool to make their eye-contact moment tangible and thus demonstrate specific phenomenological insights.

In a theoretical analysis, P3 and P4's session employs the instrument for (1-2) meshing routines, (2-4-5-6) mapping the tools which they may use to avoid the other strangers or

distractions, and (7-8-9) making sense, using the newly added body bands and possibly (9) outlining the lifeworld.

4.2. Design implications for the toolkit

Based on the co-design iterations, we prepared a first guideline describing the materials and a series of steps for the activity (Figure 7). The most important design decision that we made was to move from a detached, 'birds-eye' relation between the participants and the instrument to a more ego-centric 'view from within' approach, in which the instruments become 'engagement catalysers' (Hummels & van Dijk, 2015) rather than objects that represent people, things and their relations. In the next iteration of the toolkit, participants would find themselves literally situated inside a 'web' of tangible objects and materials that make up the instrument. This ego-centric positioning is in contrast with discrete objects placed more passively at the centre of a table with participants sitting around it (Buur et al. 2023). Being 'inside a web of things', participants are invited to have ongoing interactions with the instrument. The goal is to achieve the experience of 'having made sense', rather than create some final configuration that best represents the sense made.

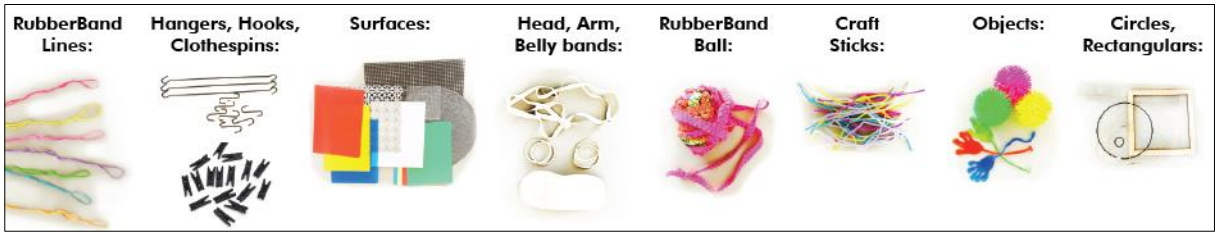


Figure 7 Materials and guidelines for the first Embodied Sensemaking Instrument prototype.

5. Iteration II: Young people mapping their energy consumption

To try out how the instrument would work with non-designers, we organised a series of workshops with young participants to see how the prototype would contribute to investigating the embodied nature of energy consumption in private homes. We sharpened the set of objects in the toolset (Figure 8). We included rubberband lines and a rubberband ball with differences in thickness, one representing the environment and the other representing the interactions of the participants.

5.1. Young people express their energy consumption

In the first two workshops, we introduced these objects without too rigid guidance and gave examples of how to interact with them. We introduced what we aimed to see in this study and how we planned to develop it together. Then, we asked the participants to think about how they consume energy in one of the daily practices of cooking, showering, or heating. We observed how the participants physicalised energy consumption and how they expressed themselves and gained awareness through the Theory Instrument.

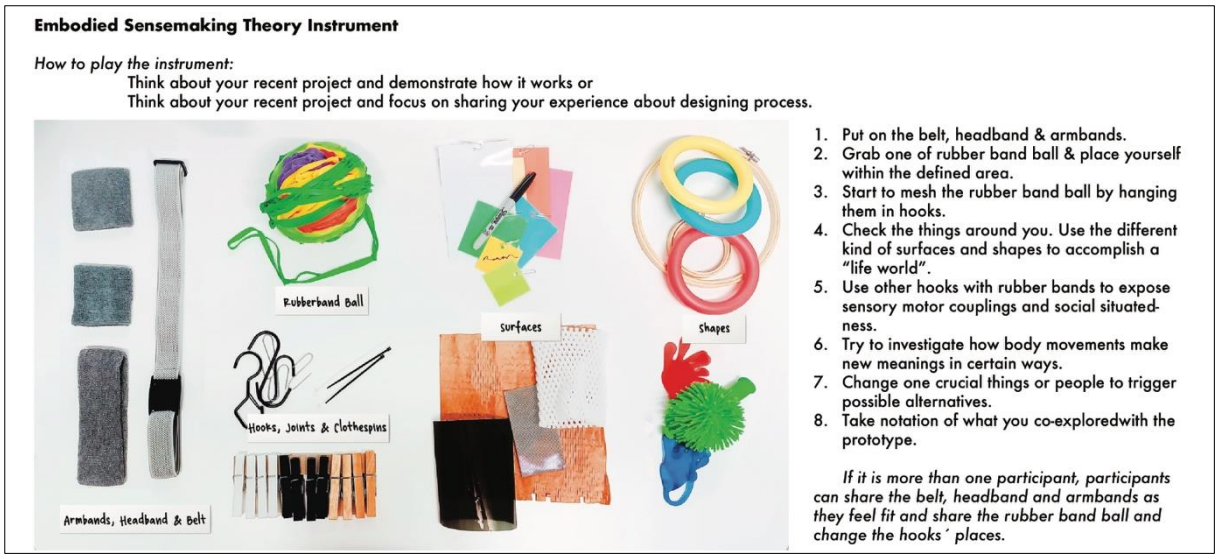


Figure 8 Refined materials of the second Embodied Sensemaking Instrument prototype.

Cooking dinner: Energy has invisible connections

The facilitator (F) hangs rubberband lines on a valet stand and introduces the materials to Participant 6 (P6). While P6 is wearing the head, arm and belly bands, F asks her to choose one of the three practices, then imagine the ball of rubberband thread as a connection tool and the lines as the environment in which she carries out the practice. She chooses *cooking* and starts to play with the lines:

P6: *When I am thinking of cooking, the first thing I think is what I would have to eat, so I go to market on my way to work then this is like when I am at work* (takes pink line), *and then this will be market* (passes the line around a chair and moves the chair to expand the area) (...) *and all of these my home* (pointed the other lines) *I do get home*.

Then she uses the ball as the route to indicate her routines (Figure 9.1):

P6: *So do I connect myself to these places?* (indicates the red line and ball line and grabs the beginning of the red line) *because now I haven't cooked all day and* (moves close to lines) *haven't even thought about it, but sometimes I think about it, and I look at the cooking app before I go. Then, if I did, it would be right in the day* (ties middle of the red line), *desperate and hungry, I would come to market* (winds around the pink market line), *and I would grab whatever was on discount and also probably just sandwiches because I am too lazy to cook.*

She picks objects to indicate sandwiches, chores and market bags, and while the plastic rings represent plates, she dedicates several plastic boards to mean household appliances. We note that she physicalises energy consumption by arranging yellow craft sticks as a lightning symbol (Figure 9.2). She also tries to measure energy usage by adding heavier, squeezable objects (Figure 9.3) on the orange line (her kitchen counter):

P6: *I used to make a roast, like pork shoulder, but I would use it for lots of dishes because it's only 99 krones for 2 kilos of meat, and I would roast it but it takes 4 hours to roast. And the roast is heavy. Like this one?* (takes the green squeezy object) *It really is weighing it down.*

(bounces on the line) *It's really weighing down the economy because I use so much electricity in the oven to make this roast, but the meat itself is very cheap, so it feels like a deal. But it costs the most of all the electricity in the house.*

She also uses craft sticks to show the invisible connections with technological devices such as the boiler, her phone and her speaker:

P6: *The boiling starts to call my attention, you know, the boiling water when it needs me. Otherwise, I can forget it, so I'm sort of disconnected but connected to it* (attached blue sticks as boiled water by colour coding them, too: Figure 9.4).



Figure 9 Embodied sensemaking of energy consumption during dinner preparation

In our theoretical analyses, P6 uses the Theory Instrument to (1) outline the lifeworld as her kitchen room, (2) map tools and skills in the kitchen, (3) mesh her daily cooking routines, and (4) make sense with invisible devices. We noticed the way of her playing with routes and craft sticks could afford more possibilities to shape, size and code the participants´ personal energy consumption and how it connects to devices.

Morning Shower: Energy consumption hidden in routines

Participant 7 (P7) chooses showering practice as a case. He recreates his showering environment (Figure 10) and demonstrates the full procedure as if he was really taking a shower:

P7: *I turn on the shower. I wait for it to be, you know, have the right temperature. I think 15 Minutes, 10 minutes, maybe 10 minutes.*

When he is done, he keeps enacting: shuts off the water, picks a round textile board as his towel, puts on imaginary clothes and opens the door (yellow line), leaving the bathroom.



Figure 10 Embodied sensemaking of the participant`s daily morning shower practice.

Like P6, P7 (1) outlines his lifeworld as a triangular bathroom with lines, (2) maps the products with boards, tiles and a round plate, (3) meshes the routines and habits with the

rubberband ball and craft sticks, (4-5) makes sense with several objects by wearing or attaching them. But differently from the previous sessions, P7, at his own initiative, also enacts a full sequence of showering - without F's prompt. We have called this (6) '*enacting phenomenology*'. In this sequence, it becomes clear how energy consumption is hidden in the routines people do.

Heaters at home: Energy has social boundaries

Participant 11 (P11) chooses heating as practice, and she starts by studying the guidelines we prepared.

P11: *I'm living in Ankara, and my city is a little cold. Even if my neighbourhood is not that cold, my family's apartment is. Because It is a big apartment. I live with my family, and I have a private room* (takes a red and green line and surrounds the chair with them: Figure 11.1)*. I don't know why it's this cold during the winter* (...) *I entered the room on that side* (sits on the chair) *here is the window and the heater system and my bed lying this way* (shows with her hand).

The facilitator (F) suggests using boards, tiles, plastic folders, textiles and papers. P11 puts the small transparent tile in the corner as a heater and explains that even though the heater works, she is feeling cold. She wraps the round textile board around her body (Figure 11.2) as her polar shirt. Green and blue paperboard become her blankets for the middle of winter.

P11: (meshes her weekend routines with the rubberband (Figure 11.3), opens the central heating system box door to check why her room is cold). *We mainly put it around 40°C during the winter. Sometimes if it's very cold, we make some minor adjustments on it. For example, we can go up to 45 or 46, And it starts at 30°C, I guess.*

In Figure 11.4, she wears different kinds of craft sticks on her head when she enters the kitchen to feel the temperature differences: her mother (the orange squeezy toy) bakes something with a delicious smell, the television opens and her sister works on her laptop (the pink squeezy toy next to the mother) while the tea machine is heating the water and makes a sound (Figure 11.5).

After this she complains about her brother's energy consumption:

P11: *my brother* (the green squeezy toy on a chair next to her room area) *is always opening his heater less and wears a t-shirt and says this house is so cold* (…) *because our rooms are next to each other* (...) *my room is always so cold because of him* (Figure 11.6).



Figure 11 Embodied sensemaking of the heating of a family house during winter.

P11 uses the material and follows the logic steps we provided to (1) outline her house as in several triangles, (2) map the heaters and other products to make heat at home, (3) mesh all daily practices while the heaters are working, (4-5) again map her sensors and skills around her face, hands and body, (5) placing her family in each room, (6) tracing some house rules and (7) enacting all phenomenological aspects.

5.2. Design implications for the toolkit

How did the instrument scaffold the participants in making sense of their energy consumption? By challenging them to trace their routines with the rubberband ball it becomes apparent that energy consumption is tied to embodied routines. Similarly, the challenge to choose and place objects for community members helps underline the social nature of energy consumption. More interestingly, perhaps, is the finding that energy consumption is influenced by 'invisible connections' to devices. This comes about through the weaving of lines.

Although the participants generally found the variety of materials positive, and the affordances enticing, they had difficulty mastering all the materials and expressing abstract concepts physically. The first two sessions on energy consumption lead us to develop a more precise guideline of logic steps and a more narrow definition of material actions, Figure 12. We used this consistently throughout the next 5 sessions to confirm that it works.

Throughout the sessions, we observed that – although the participants repeatedly asked what each object 'means' – it was really the actions that the participants did with the objects that carried conceptual meaning. To that end, we have singled out the 'object actions' and suggested a set of 'instrument terms' in Figure 13.

Importantly, and in contrast to Theory Instruments designed previously, having people experience the content of this theory does not work very well by re-presenting the theory in the form of *physical metaphors*. When it comes to understanding the meaning of aspects like action-perception coupling, lifeworld, and social situatedness, it helps if one is actually *acting,* if one feels being situated in a lifeworld, rather than using tangible objects that represent (symbolize) such concepts. This boils down to saying that one needs to literally put one's body into the sensemaking activity, rather than working with objects in a more detached manner.

THE LOGICS OF
EMBODIED SENSEMAKING INSTRUMENT



OUTLINING LIFEWORLD

- Choose one of the situated practise above.
- Place yourself where you are practicing as in your house, office, school, or shop.
- Map your environment according to the colors of lines& the distance and width you want.
- Use hooks or clips if you need to hang Lines or use elasticity feature of rubberbands.
- Show the most electricity usage area.





MAPPING TOOLS AND SKILLS/MESHING ROUTINES AND HABITS

- Shape surfaces as objects, products you use daily.
- Place them in their place in real life as the most appropriate way with hooks or clips.
- Pay attention to the colors, shapes textures, transparency of materials and explain why...
- Put the hairbands on your hands and feets, wear the headband above your head and put the belt around your belly with a hook.
- Tie the rubberband ball to your belt and ready to carry on your hand.
- Start to link, tie, or connect from the first place of your practise to the last.
- Think and share what you are doing& how you are doing the practise; Pull, Stretch, Open, Wrap...
- Shape pipecleaners as devices or systems you use daily.
- Clip on them on your Headbans, Hairbands or Belt & Lines by paying attention to the color code.
- Show electricity consumptions on the products



PLACING COMMUNITY/TRACING NORMS AND RULES

- Hang random objects on Lines and Ball with hooks or clips or embed between their knots as people, community around you.
- Pay attention to the colors, shapes, textures, elasticity and explain why...
- Hang or embed circles and rectangulars on Lines and Ball as Challenges, Culture, Background, Ethics, Religions
- Pay attention to the colors, shapes, thickness and explain why...
- Show the electricity usage through people.



EMBODIED SENSEMAKING/ENACTING PHENOMONOLOGY

- Focus on where and how you and other peole are using energy.
- Enact all steps of practise from the beginning till the end with full-bodily engaged.
- Share your reflections and how you are feeling about your energy consumption.

Figure 12 Guidelines of how to 'play' the Embodied Sensemaking Instrument.

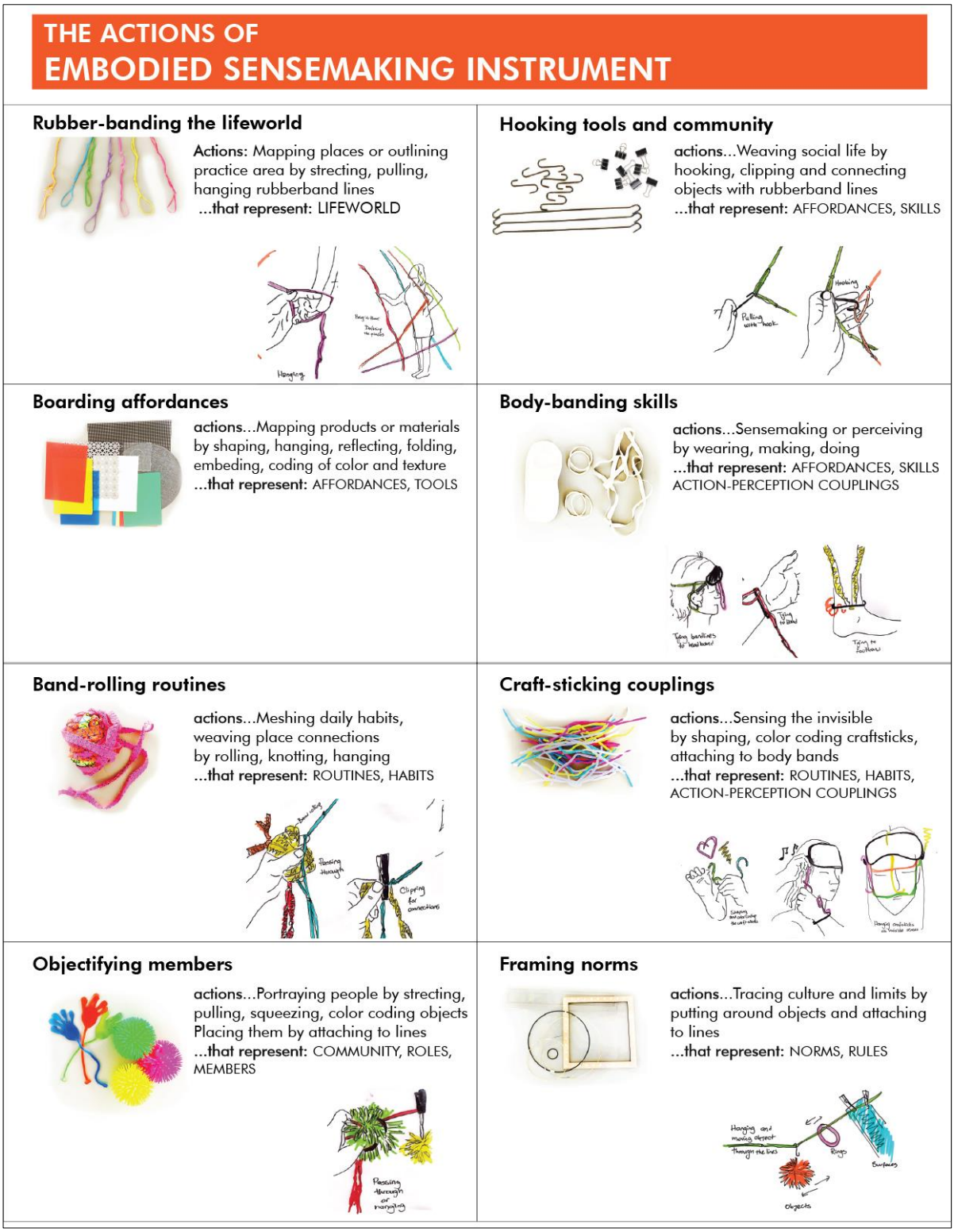


Figure 13 An overview of 'object actions' that carry theoretical aspects in the Embodied Sensemaking Instrument

6. Discussion

Much of the previous work with Theory Instruments relies on meaning attribution to specific tangible elements that represent theoretical concepts (Buur et al., 2022). In this instrument, the conceptual understandings were attributed less to the specific objects than to specific experiences.

We observed two distinct ways in which participants interacted with the Theory Instrument and thus made sense of both the theory itself and, through it, the experience they were exploring: (a) the staging of the experiments and (b) the specific actions afforded by the instrument. The first, staging, refers to how the experience overall was staged and narrativised. This led, midway through our experiments, to the emergence of distinct stages in participants' narratives as they explored the Theory Instrument, from the outlining of lifeworld, mapping of skills and habits, investigation of norms and rules, and finally, to the enactment. In our later iterations of the task, we formalised these stages so that each corresponds to a specific set of theoretical concepts that the participants experienced through their interactions with the instrument.

The second refers, more closely, to the specific set of bodily actions that the participants undertook in their interactions with the Theory Instrument. We realised that it was the actions rather than the objects that represented the various elements of the theory, in line with sensorimotor accounts of embodied sensemaking (van Dijk, 2018; Hummels & van Dijk, 2015; Wilde et al., 2017; Kirsh, 2013; Schaper & Pares, 2021). As demonstrated, it was the full bodily action and sense of hanging, pulling, etc., that helped the participants make sense of the theory and, through that, the experience they were exploring. By doing that, our participants were experiencing theoretical concepts through embodied engagement.

Over the course of a workshop, these two perspectives are not mutually exclusive; participants will switch between objective perspectives and personal ego-centric perspectives in dealing with the instruments. Like previous Theory Instruments this one invites action, ‘playing around with,’ fidgeting and experimenting with the material objects, and it also leads to things being named and given explicit representational attributions. But we see it as a shift in focus that the current toolkit makes the user attend more to ‘doing things with things’ and experiencing how ‘that makes sense’ rather than attributing meaning to an object through talking about that object.

By developing the facilitator role from ‘neutral observer’ to ‘active co-actor’, we realised that the instrument is best played socially with more than one participant. When the facilitator intervenes in the process, she leads participants to improvise their story together. These unpredictable acts enrich the performance and outcomes.

It is an open question whether the basic processes of embodied sensemaking that we surface are culturally dependent, or a fundamental feature of human being. If the body is crucial, then different bodies will generate different sensemaking. But all these different bodies may still display general patterns, such as a switching between representational and non-representational forms of interaction with the material surroundings. It would be very

interesting to investigate this, and to do so in a way that not already biases the study with our own (Western/European) ways of sensemaking.

We believe that the Embodied Sensemaking Instrument paves the way for a new way of integrating a theoretical perspective in design work, in particular in investigating how people relate to consumption and sustainability. While 'playing the instrument' to reimagine futures still needs work to stabilise the activity for actual practice, our research shows a promising design space for more Theory Instruments to be developed.

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